

■ 2.1.4 Applying Functions by Name

The “name” f of a function, like everything else in *Mathematica*, is just an expression. As a result, you can have functions that take as arguments the *names* of other functions.

`Nest[f, x, n]` takes the “name” f of a function, and applies the function n times to x .

```
In[1]:= Nest[f, x, 5]
Out[1]= f[f[f[f[f[x]]]]]
```

`Apply` lets you apply a function to a list of arguments.

```
In[2]:= Apply[f, {a, b, c}]
Out[2]= f[a, b, c]
```

This gives the sum of the elements in a list.

```
In[3]:= Apply[Plus, {a, b, c}]
Out[3]= a + b + c
```

<code>Apply[f, list]</code>	apply the function f to the elements in a list
<code>Nest[f, x, n]</code>	apply the function f nested n times to x

Applying named functions to expressions.